

Reviewer Pack — Minimal Rank of Quasi-Linear Representations Bounds DPLL Ref...

Entry 4494bcbb4264 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Quasi-Linear Representations Bounds DPLL Refutation Size

Entry ID: 4494bcbb4264 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-27 13:39:35 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Representation Theory (Quasi-linear Representations)

Field B (complexity object): Complexity Theory: DPLL Search Trees

Statement:

{‘st1’: ‘For any given boolean function f , the minimal rank of its quasi-linear representation $\rho(f)$ is strictly greater than the depth of the minimum-size DPLL refutation for f .’, ‘st2’: ‘That is, there exists a constant $c > 0$ such that $\rho(f) \geq c * d(f)$, where $d(f)$ is the depth of the smallest DPLL tree refuting f .’, ‘st3’: ‘Furthermore, this relationship holds modulo linear equivalence classes of representations.’}

Rationale (proposer’s reasoning):

{‘r1’: ‘Quasi-linear representations offer a structured way to encode boolean functions, which may reveal underlying complexities that are not evident in classical representations.’, ‘r2’: ‘The study of these representations could potentially expose new insights into the complexity of boolean functions and their refutation processes.’, ‘r3’: ‘If such an invariant is indeed useful for DPLL search trees, it would provide a novel approach to understanding complexity separations.’}

Taxonomy category: GCT_DET_PERM (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 6549f100d3c633a0

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a given boolean function, if the ratio between its quasi-linear representation rank and the minimum DPLL refutation depth exceeds a constant c for at least 30 random instances with size $n \leq 40$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	SAFE	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 3 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - algebraic representation theory quasi-linear representations AND DPLL refutation size - DPLL search trees AND minimal rank quasi-linear representations - linear equivalence classes quasi-linear representations AND complexity theory DPLL

Top relevant hits considered: - [<http://arxiv.org/abs/math/0701775v1>] Global Low Regularity Solutions of Quasi-linear Wave Equations - [<http://arxiv.org/abs/1208.0993v3>] Equivalence Classes of Colorings - [<http://arxiv.org/abs/1611.00827v2>] Geometric complexity theory and matrix powering

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=124, elapsed=240.3s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def dpll_refutation_depth(f):
        n = len(f)
        if n == 1:
            return 1
        depth = 0
        for i in range(n):
            sub_f = f[:]
            sub_f[i] = 1 - sub_f[i]
            depth = max(depth, 1 + dpll_refutation_depth(sub_f))
        return depth

    def quasi_linear_representation(f):
        n = len(f)
        if n == 1:
            return [f[0]]
        representation = []
        for i in range(n):
```

```

        sub_f = f[:]
        sub_f[i] = 1 - sub_f[i]
        representation.append(sub_f)
    return representation

def rank(representation):
    n = len(representation)
    if n == 1:
        return 1
    max_rank = 0
    for i in range(n):
        sub_rep = representation[:]
        sub_rep.pop(i)
        max_rank = max(max_rank, rank(sub_rep))
    return max_rank + 1

def linear_equivalence(r1, r2):
    n = len(r1)
    if n != len(r2):
        return False
    for i in range(n):
        if not all(x == y for x, y in zip(r1[i], r2[i])):
            return False
    return True

def find_counterexample(f, d):
    representation = quasi_linear_representation(f)
    rank_rep = rank(representation)
    if rank_rep < d:
        return f"Counterexample found: f={f}, d={d}, rank_rep={rank_rep}"
    return ""

n = random.randint(5, 40)
f = generate_boolean_function(n)
d = dpll_refutation_depth(f)
representation = quasi_linear_representation(f)
rank_rep = rank(representation)

if rank_rep < d:
    counterexample = find_counterexample(f, d)
else:
    counterexample = ""

return {
    "metric_name": "rank_over_d",
    "metric_value": rank_rep / d,
    "instances_tested": 1,
    "conjecture_holds": rank_rep >= d,
    "counterexample": counterexample
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:

```

```

    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_rank_over_d = sum(r["metric_value"] for r in results) / len(results)
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_rank_over_d} std=0.0 support_fraction=1.0")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_rank_over_d} std=0.0 support_fraction={support_fraction}")
else:
    first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means it was unable to complete the required number of instances to evaluate the conjecture. | next: Re-run the test with increased time limits or consider alternative methods for verifying the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12905
2	preregistration	ollama_remote	glm4:latest	0	0	6518
3	novelty	ollama_remote	glm4:latest	0	0	4698
4	novelty	ollama_remote	glm4:latest	0	0	6620
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	41544
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11511
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10271
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9619
9	judge	ollama_remote	glm4:latest	0	0	28406

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 132091 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/4494bcbb4264.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/4494bcbb4264.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/4494bcbb4264.tar.gz (if generated)